

Evaluating Measurement Quality

Fundamentals of Data Collection II

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Why Evaluate Measurement Quality?

- Despite following best practices for research design, the data that are collected – the individual observations – are not necessarily accurate
- This is measurement error
 - usually defined as the difference between observed and true values
- True value is usually unknowable
 - if known, no reason to collect data
- but numerous proxies for the true value have been used creatively over the years

Outline

- Direct Measures/Gold Standards
 - Administrative Records
 - Passively Measured (e.g., Sensor) Data
 - Scenarios/Vignettes
 - Indirect Measures
 - Response change between waves
 - Benchmark surveys (e.g., gov't telephone surveys)
 - Response Distributions
 - Correlational Measures
 - Concurrent/predictive validity
 - Surveys as gold standard in evaluating organic data
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❖ most examples presented previously in SM 621 or SM 622

Validating Responses with Administrative Records

- Records of *Rs'* attributes, behaviors, etc., may exist that can validate their self-reports in methodological studies
- Different in spirit from “record linkage”
 - which is usually done to avoid asking survey *Qs* if relevant data already exist or to expand set of analytic *Qs* that can be answered
- But involves many of same steps and challenges
 - e.g., identifying *Rs'* records despite imperfect name match

Administrative Records, Example 1

- More variation in responses between *lwers* (higher ρ_{int}) is typically assumed to reflect differences in how *lwers* conduct interviews
 - i.e., more variance between *lwers* => more *measurement error*
- West and Olson (2010) point out that high ρ_{int} might be due not only to measurement error but also to differential *nonresponse* error
 - i.e., different true values in the subsamples interviewed by different *lwers* is possible (despite interpenetrated design) due to different recruitment approaches and outcomes
 - different types of *Rs* are selected – i.e., agree to be interviewed – by different *lwers*, presumably due to differences in recruitment strategies and abilities
- West and Olson (2010) used administrative records to quantify relative contributions to ρ_{int} of nonresponse (recruiting) and measurement (interviewing)

Admin. Records, Example 1 (cont'd)

- Wisconsin Divorce Study (WDS) sampled potential participants from frame of 733 divorce certificates from four counties in WI; design “approximates” interpenetration
- Divorce certificates included “official information” also collected in WDS
 - Divorce dates, marriage dates, birth dates, etc.
 - assumed to be error-free, although also self-reported
- Divorce records enable evaluation of error due to measurement and nonresponse
 - Measurement error: discrepancy between survey responses and values in records
 - Nonresponse error: differences between values in records for those who did and did not participate
 - Differences between *Int*ers in either type of error will be reflected in higher ρ_{int}
- Evidence of both types of error in different questions:
 - Two Qs produced interviewer effects (ρ_{int} significantly > 0): *Age at Marriage* and *Age at Divorce*
 - Based on records, effect for *Age at Marriage* due to measurement error, but effect for *Age at Divorce* due to nonresponse error

Administrative Records, Example 2

- Debate over pros and cons of standardized interviewing (SI) concerns impact of misunderstanding *Qs* on response accuracy
 - Holding *Q* wording constant assumed to promote comparability of responses
 - but blocks *conversational grounding*, i.e., *I*wer and *R* making sure on same page
- Conversational Interviewing (CI) proposed as method of improving response accuracy by improving – standardizing – *Rs'* comprehension of *Qs*
 - *I*wers able to assure *R* understands *Q* as intended (to “ground meaning”)
 - but could increase ρ_{int} if CI implemented differently by different *I*wers
- West et al. (2017) assessed these concerns by combining response accuracy and interviewer variance (IIC*) in Mean Square Error (MSE)
 - calculated response accuracy – actually inaccuracy (bias) -- by computing discrepancy between responses and admin (employment) records

* IIC = Intra-Interviewer Correlation; same as ρ_{int}

Admin. Records, Example 2 (cont'd)

- West et al. (2017) report that, based on admin records, CI led to significantly higher accuracy than SI for the one Q for which admin data was available
- For this Q , more *lower*-variance (higher IIC) for CI than SI
- So more accurate answers but more *lower*-variance in CI – which really reflects data quality under CI?
- When combined into MSE , * increased IIC for CI was outweighed by higher response accuracy (match with records), suggesting greater overall quality for CI than SI

* MSE is squared bias (discrepancy between records and responses) plus variance (including IIC)

Challenges Using Admin. Records for Validation

- Unless survey Qs are written with existing records in mind, unlikely that useful records will be available
- Data management and data manipulation
- Timeliness of admin. data
 - Available records may lag survey data by several years
- Consent bias
 - response accuracy may differ for consenters and non-consenters
- Inaccuracies exist in records
 - especially if records created from self-reports, e.g., tax records

Sensor Data

- Data passively collected with sensors native to smartphone or with wearable sensors (e.g., FitBits) are sometimes used to validate self-reports
 - Objective measures, e.g., steps or geo coordinates
 - no forgetting or lapse in attention by participant
- Scherpenzeel (2018) tracked participants' travel behavior with app and asked them to self-report their trips
 - App tracked trips using GPS, cell tower triangulation, and Wi-Fi
 - App registered about 7% more trips per day than self-reported trips suggesting self-reports underestimate actual travel
- But sensor (GPS) data not always perfectly accurate
 - intermittent measurement in “urban canyons”
 - Participants deleted some passively detected trips from the list of passively measured trips suggesting some “false alarms” in sensor data
 - 14% indicated “certainly not true” that app “detected my trips correctly”

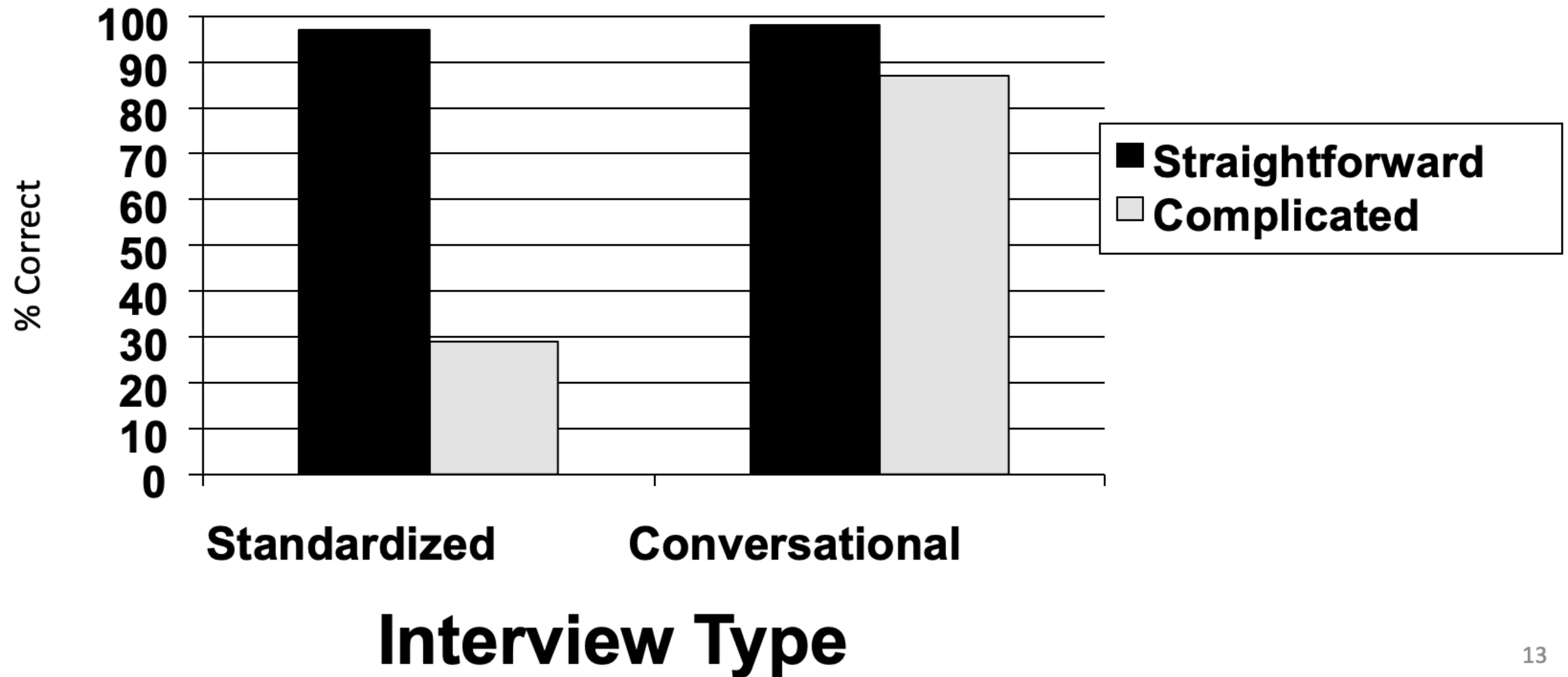
Scenarios

- Instead of asking *Rs* to answer about their own lives, they can be asked to respond to fictional scenarios for which true value known
 - because researchers create the scenarios with the correct answer in mind
- This approach has been used to compare response accuracy in SI and CI
- In CI, *Iwers* can communicate intended interpretation to *Rs* based on official definitions
- But in SI, *Iwers* can only use neutral probes like “whatever it means to you” or “let me repeat the question”

Scenarios (2)

- Example:
 - Q asks about any purchases for household furniture
 - scenario is a receipt for purchase of a lamp
 - definition excludes lamps and lighting fixtures from “household furniture”
 - correct answer – true value – is “no”
- Schober & Conrad (1997) used this approach for 12 Qs, creating a straightforward and complicated (ambiguous) scenario for each
 - Main research question was can CI resolve ambiguity in complicated scenarios (like receipt for lamp purchase) and improve response accuracy

Response Accuracy (Agreement with Definition)



Challenges Using Scenarios for Validation

- Ecological validity:
 - response processes may differ from those in actual interview so results may not generalize
 - although in SI-CI comparisons, results are consistent across other validation measures
- Mostly applied to comprehension processes; not typically used to validate measurement where recall, estimation, and response selection are at issue

Response Change Across Interviews

- In typical reinterview studies* consistency or reliability across interviews is taken as evidence of high-quality responses
 - If administration is identical across first interview and re-interview, minimal change between waves indicates high quality data
 - Substantial change between interview and re-interview is a red flag
- But logic can be reversed for assessing measurement quality:
 - if experimental treatment is administered in reinterview that is intended to improve response quality, change in response (in expected direction) indicates improvement, all else being equal

* e.g., McGovern & Bushery, 1999

Response Change, Example

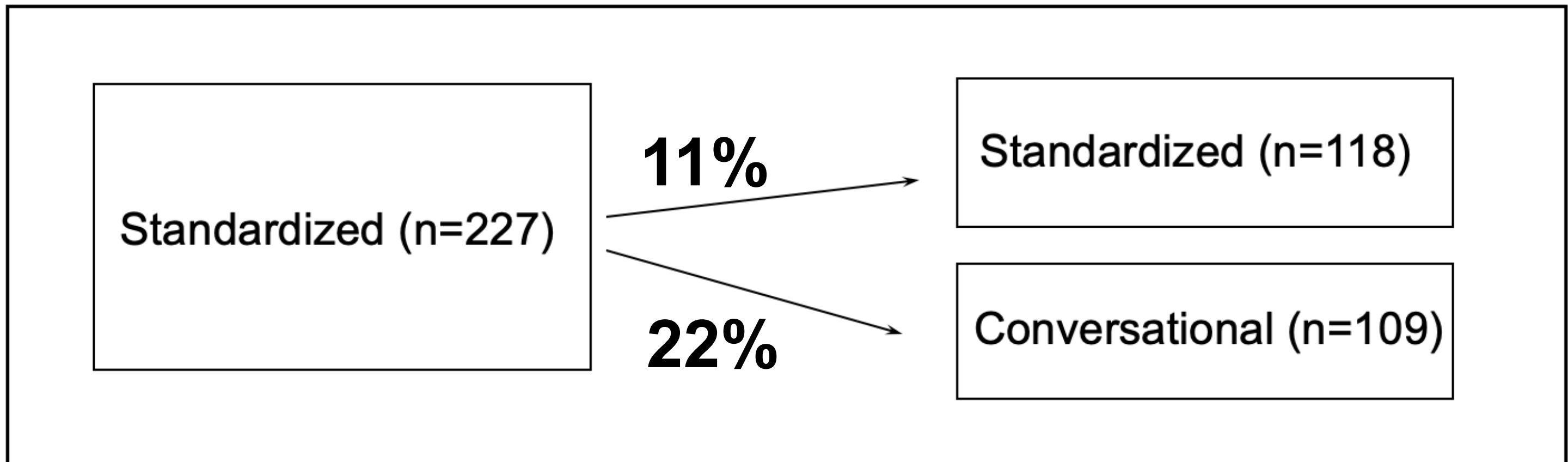
- Conrad & Schober (2000) asked *Qs* via *SI* in Wave 1 but in second wave re-asked *Qs* via *CI* for half *Rs*, *SI* for other half *Rs*
- If *CI* improves comprehension (and, presumably, response accuracy) in second interview, we expect more response change between wave 1 answers (*SI*) and wave 2 answers when *Qs* re-asked in *CI* than *SI*
 - Second administration should repair wave 1 misconceptions

Results: Response Change

- More responses changed when second interview was conversational than standardized

INTERVIEW 1

INTERVIEW 2



Benchmarks

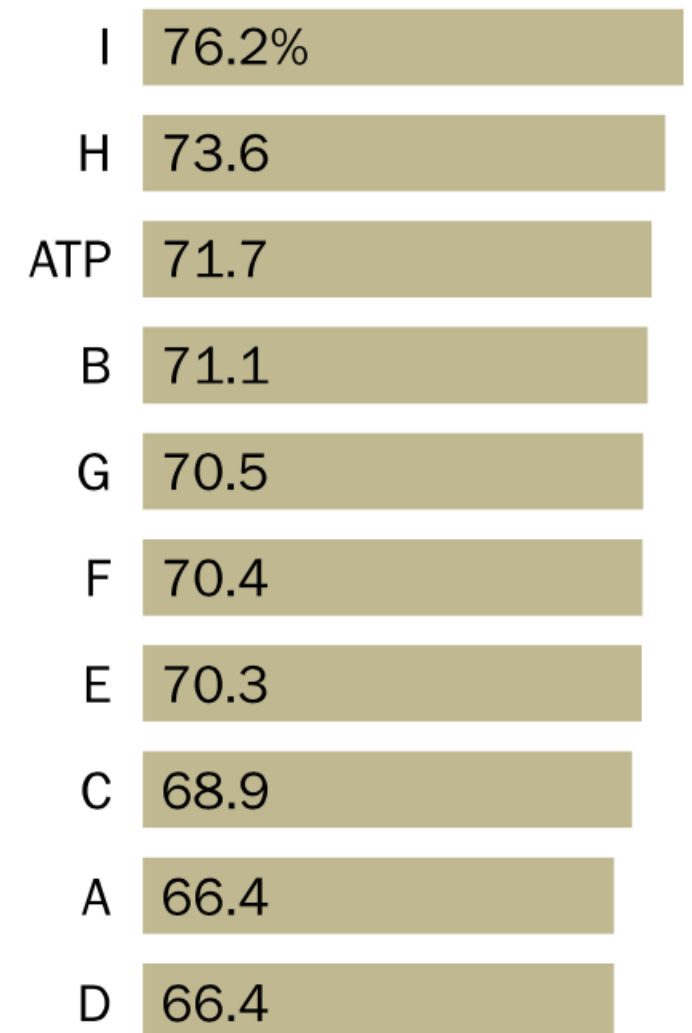
- Studies that evaluate impact of data collection mode and sample source on survey estimates sometimes treat estimates from trusted source as true value, i.e., benchmark
 - e.g., government surveys: high response rates and considerable expense and care is invested in their quality
- Accuracy (or its inverse, bias) is deviation between estimates (e.g., mean, %) based on a survey Q and benchmark values
- Benchmarks are aggregate gold standards, i.e., do not provide validation for individual respondents' answers

Benchmarks

- A Pew study (Kennedy, et al., 2016) compared 20 survey estimates from 9 non-probability online samples to benchmark population estimates
 - Benchmarks derived from “high quality federal sources [including] national surveys...” plus some admin records
 - Topics include smoking, health care coverage, income, participation in civic or recreational organizations, voting, household composition, internet usage
- Also compared American Trends Panel (ATP), Pew’s online probability panel to benchmarks

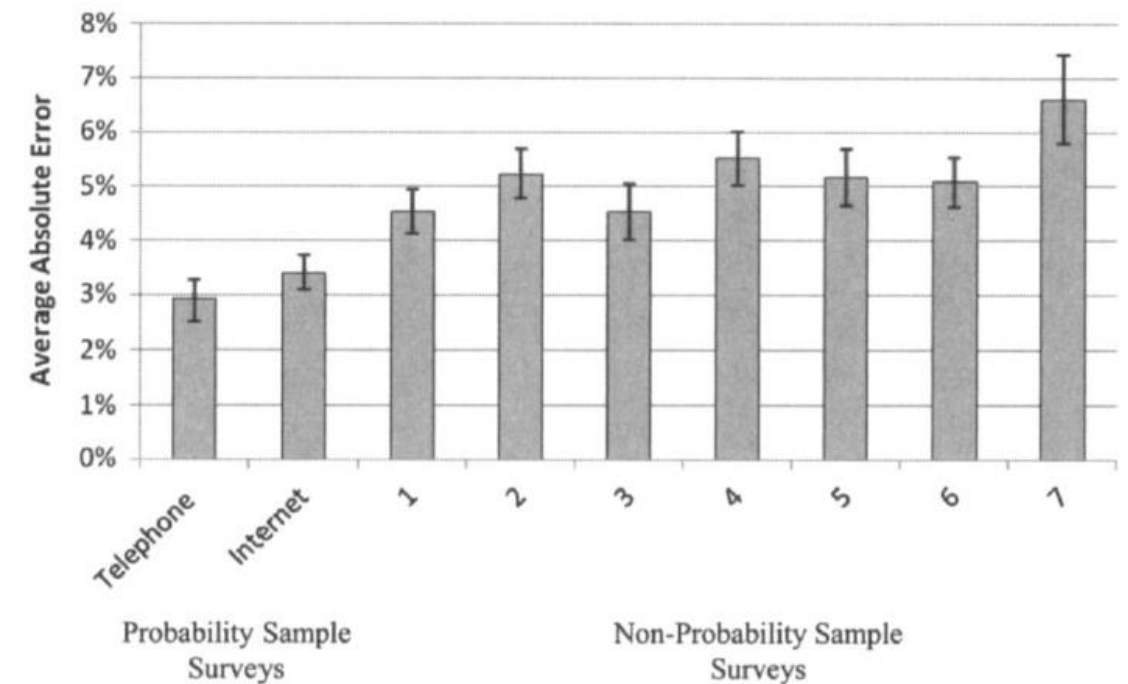
Average % correctly classified in regressions

How well regression models from online samples predict outcomes on benchmark samples (average across four outcomes)



Benchmark Surveys (2)

- Yeager et al. (2011) compared accuracy (% absolute error) for 7 non-probability web surveys, one probability web survey and a probability telephone survey
- “Error” was the deviation between responses from each survey and
 1. estimates from large US gov. surveys with response rates > 70%, and
 2. admin. data from US State Department (# passports), US Federal Highway Admin. (# drivers licenses)



- What is an example of a benchmark that does not rely on surveys or administrative records?
- What are some concerns/caveats about the use of benchmarks?

Response Distributions

- There may be theoretical reasons to expect responses from two experimental conditions to differ
 - e.g., a larger mean for a variable in one condition than the other
- To the extent the expected pattern is observed, this may be taken as support for the hypothesis
- Sometimes, the difference is not predicted to be in one direction or the other – just different
- In other cases, direction is predicted (e.g., Condition A > Condition B) and results in one direction assumed to be closer to the true value

Response Distributions, Example 1

- *Rs'* reports of behavioral frequency is influenced by the instances of the behavior that come to mind
- Couper, al. (2004) reasoned that content of images in web surveys can affect what comes to mind and, thus, the frequencies that *Rs* report
- Presented image of a relatively rare (low frequency) instance or a relatively common (high frequency) image of six behaviors
 - e.g., for *sporting events in past year*, low frequency (LF) image of a professional baseball stadium and high frequency (HF) image pictured an amateur (kids') game
 - Predicted pattern of reported frequency: $LF < HF$
 - Observed this pattern of mean reported frequencies for most of the six behaviors

Response Distributions, Example 2

- If *Rs* exposed to a definition that affects understanding of *Qs*, may lead to a more *inclusive* or more *exclusive* interpretation
 - Inclusive: “Income” defined to *include* overtime pay
 - Exclusive: “Moving expenses” defined to *exclude* buying food for friends who helped you move
- West et al., (2017) used this approach to compare accuracy of Conversational Interviews (CI) and Standardized Interviews (SI)
 - Reported that CI led to larger estimates than SI for 2 *Qs* with inclusive def’s. and smaller estimates than SI for 4 *Qs* with exclusive def’s. and
- Concluded CI estimates are more accurate because differences are in predicted direction

Disclosure

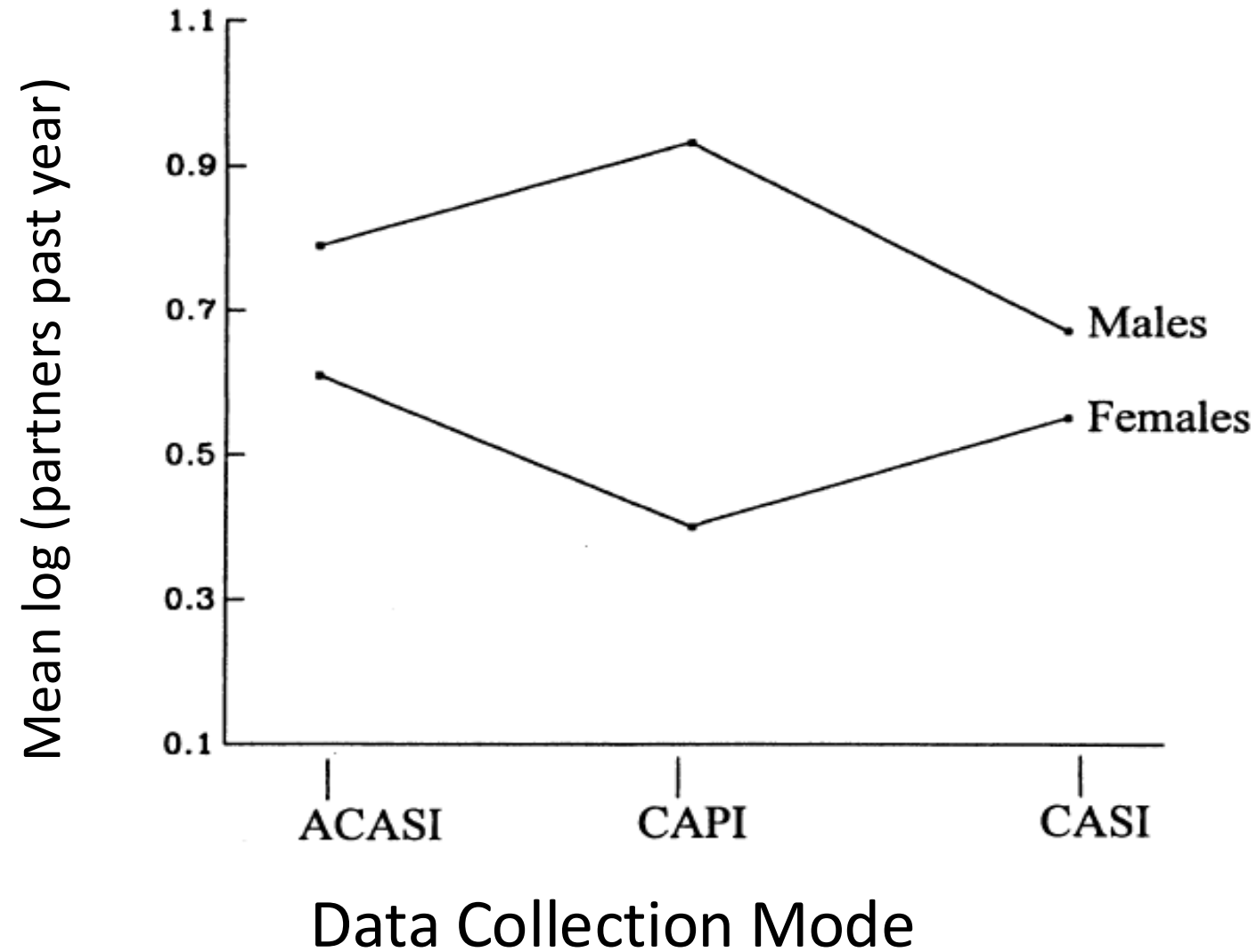
- Reporting sensitive or stigmatized info generally makes people uncomfortable
- So, more disclosure usually assumed to be more candid
 - Typically, no reason to intentionally overreport stigmatized behavior, placing oneself in a negative light
- Long tradition of attempting to increase disclosure of sensitive behavior through revised question wording (e.g., forgiving wording) but most successful method is self-administration
- To the extent that self-administered modes promote more disclosure, results are assumed to be more accurate
 - i.e., less driven by social desirability

Disclosure, Example 1

- Tourangeau & Smith (1996) compared disclosure between several *lwer*- and self-administered modes
 - For number lifetime sex partners (excluding same sex), males assumed to overestimate and females to underestimate
 - Self-administration (CASI and ACASI) assumed to increase disclosure (honest responses)
 - Compared to in-person interviews, self-admin should lead to fewer partners reported by males, more partners reported by females
 - If observed, assumed to be higher quality data

Reported sex partners by gender and mode

(Tourangeau & Smith, 1996)



Disclosure, Example 2 (Direct Measure)

- Very few studies have verified that patterns we characterize as indicating more disclosure do in fact represent more disclosure (honest responses)
- Best known study to do this is Kreuter, Presser & Tourangeau (2008)
 - UMD alums randomly assigned to telephone, IVR or web survey about undergraduate academic experience
 - Sensitive questions included *ever having received a D or F or GPA below 2.5*
 - Consistent with logic of most studies of mode and sensitive topics, *Rs* disclosed more sensitive info, e.g., “yes” responses, when they self administered Qs : Web > IVR > Phone
 - Compared to Registrar’s records, reverse for “False negatives:” Web < IVR < Phone
- “More is better*” logic of disclosure supported by these results
- Of course, there are no records for most sensitive behaviors

*more socially undesirable answers => higher quality data

Concurrent and Predictive Validity

- When no gold standard available, possible to assess response quality by correlating answers to target *Qs* with answers to other related *Qs* or events
 - Other *Q* in same interview: *concurrent validity*
 - Other *Q* in future interview or outcome of public event, e.g., election: *predictive validity*
- Chang & Krosnick (2009) used this approach to compare measurement quality between surveys conducted with (1) a probability sample interviewed by phone, (2) an online non-probability sample, and (3) an online probability sample
- Concurrent validity measured as the extent to which variables like Clinton job approval predict (correlated with) reported likelihood *R* will vote for Gore (then Clinton's VP) than Bush (both responses collected in the pre-election survey)
- Predictive Validity measured extent to which Clinton job approval (pre-election) also predicted post-election reports of voting for Gore over Bush

Concurrent and Predictive Validity (Chang & Krosnick)

- Concurrent validity: online probability > telephone > online non-probability
- Predictive validity: online probability > telephone > online non-probability
- Leads to conclusion that, at least for political data, online probability panel produces best quality data

Organic Data: Survey Data as Gold Standard

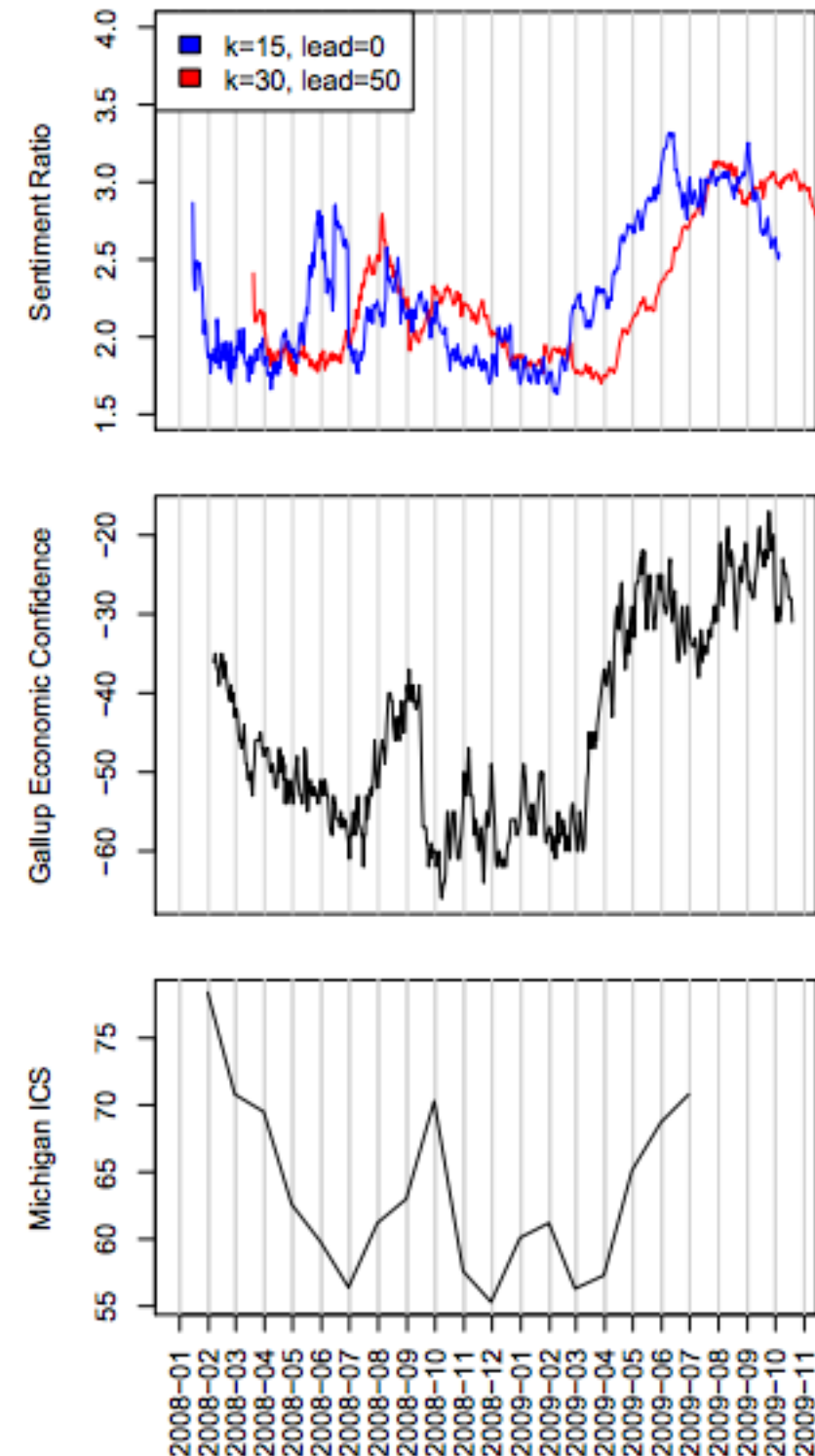
- Social media posts as data for conducting social research
- Alignment with survey data, especially time series, considered evidence that data derived from social media are credible source of information about social phenomena
 - Social media created by non-probability sample of the public
 - Could easily not align with survey data, from either probability or non-probability sample
 - So, when the two data sources do align, may be worth taking seriously
- Yet, inaccuracies in survey data (TSE framework) well known

Measuring Consumer Sentiment With Tweets about Jobs

- *Sentiment* of tweets containing the word *jobs* on each day correlated with survey-based measures of consumer confidence in 2008-2009 (O'Connor et al., 2010)
 - a. Sentiment (-1, 0, +1) assigned to each word by OpinionFinder; a tweet is positive if contains a positive word, negative if contains a negative word
 - b. Daily sentiment = # positive tweets / # negative tweets
- Compared daily sentiment ratio of tweets to two survey-based measures of consumer sentiment:
 1. Gallup's Economic Confidence Index (2 questions)
 2. Index of Consumer Sentiment (ICS) from U. Michigan Surveys of Consumers (5 questions)
 - Both measure some version of "will economy get better, get worse, or stay same?"

O'Connor, et al. (2010):

- Social media-series (top) show dropping (less positive) sentiment (with a late spike) across 2008 and upturn across 2009 ... much like the two survey-based time-series (middle and bottom)
- Authors tested impact of different smoothing intervals and lags (or leads) on correlation between social media- and survey-series
- With 15-day smoothing and 20-day lead, Gallup-Twitter $r = .73$.
- With 30-day smoothing, and 50-day lead, ICS-Twitter $r = .64$



Conclusions

- True value is real but often not knowable
 - Most social scientists believe it is “out there”
 - But very hard to operationalize
- Administrative data are a straightforward proxy of true value
 - Not available for most survey variables
 - Sometimes only available in the aggregate, e.g., benchmark data
- Sensor data are objective but not free of measurement error
 - Not helpful in confirming accuracy of opinion measurement

Conclusions (2)

- Fictional scenarios afford researchers a high degree of control over true value, especially in conjunction with official definitions
 - Challenges include ecological validity
 - Better for assessing comprehension than other response difficulties
- Response change over time/waves, differences in response distributions, correlational measures reflect quality *indirectly*
 - But if consistent with theoretically-based predictions, can help establish data quality

Conclusions (3)

- Social desirability of answers seems to be effective measure of *Rs'* honesty, and varies with survey mode
 - Variation in socially desirable responding corresponds to administrative data in the few studies that have tested this
 - Supports the “more is better” assumption, so the fact that most sensitive behaviors are not recorded, may be okay
- Survey data as true value for organic data
 - Because it is *designed*, survey data is more likely accurate than data created for purposes other than measurement
 - But overlooks the error inherent in survey data that methodologists and researchers work hard to minimize